

CODAMETRIX · HACKATHON 2026

Clinical Note Classification

Routing patient notes to the right specialty with NLP

Cardiology

Neurology

Orthopedics

Gastroenterology

Other

June 13, 2026 · Boston, MA

Read a clinical note, predict its specialty

Mission

Build an NLP model that reads a short clinical note and predicts which of five specialties it belongs to.

Dataset

~14,400 labeled notes (Medical Abstracts TC corpus) + orthopedic abstracts sourced from PubMed.

Scoring

Performance on a hidden test set - generalization over memorization.

Case 1

Single-Class

One label per note (argmax). This is the submission.

Case 2

Multi-Class Probabilities

A calibrated score per specialty. Surfaces confidence and overlap.

What the data really is - and how we shaped it

The provided data is the **Medical Abstracts TC corpus** (Neoplasms, Digestive, Nervous, Cardiovascular, General) - not the poster's five names. We remapped it to the target categories:

Source class	Target category	Notes
Cardiovascular	Cardiology	3,051
Nervous	Neurology	1,925
Digestive	Gastroenterology	1,494
Neoplasms + General	Other (oncology folded in)	7,968
PubMed (MeSH C05)	Orthopedics (newly sourced)	2,495

Why this matters

- The poster lists Orthopedics, but the data has none - so we sourced it.
- Oncology has no target slot, so it folds into Other.
- Other becomes a broad catch-all - the hardest class.

Metrics chosen through a healthcare lens

Per-class Recall

Missing a case is the dangerous error - catch every specialty.

Per-class Precision

Wrong routing burdens the wrong department.

Macro-F1

Weights the rare classes equally - no hiding behind the majority.

Confusion Matrix

Shows which specialties get confused - dangerous vs benign.

Calibration / ROC-AUC

For the probability view: are the scores trustworthy?

Risk-Coverage

Lets the model abstain on garbage instead of guessing.

In healthcare, a confident wrong answer is worse than an uncertain one.

Three model families, one honest comparison

CLASSICAL

TF-IDF + Logistic Regression

- Vocabulary features, class-balanced
- Trains in seconds on CPU
- Interpretable, hard to overfit
- Shipped submission

TRANSFORMER

PubMedBERT (fine-tuned)

- Reads notes in context (semantic)
- Pretrained on PubMed abstracts
- Fine-tuned on MPS (M4)
- Understands negation, synonyms

LLM

Claude (few-shot)

- No training - 6 in-context examples
- Pure semantic reasoning
- Best for style shift / transcripts
- Robust on unseen phrasing

One label per note (argmax)

0.70

Accuracy

0.71

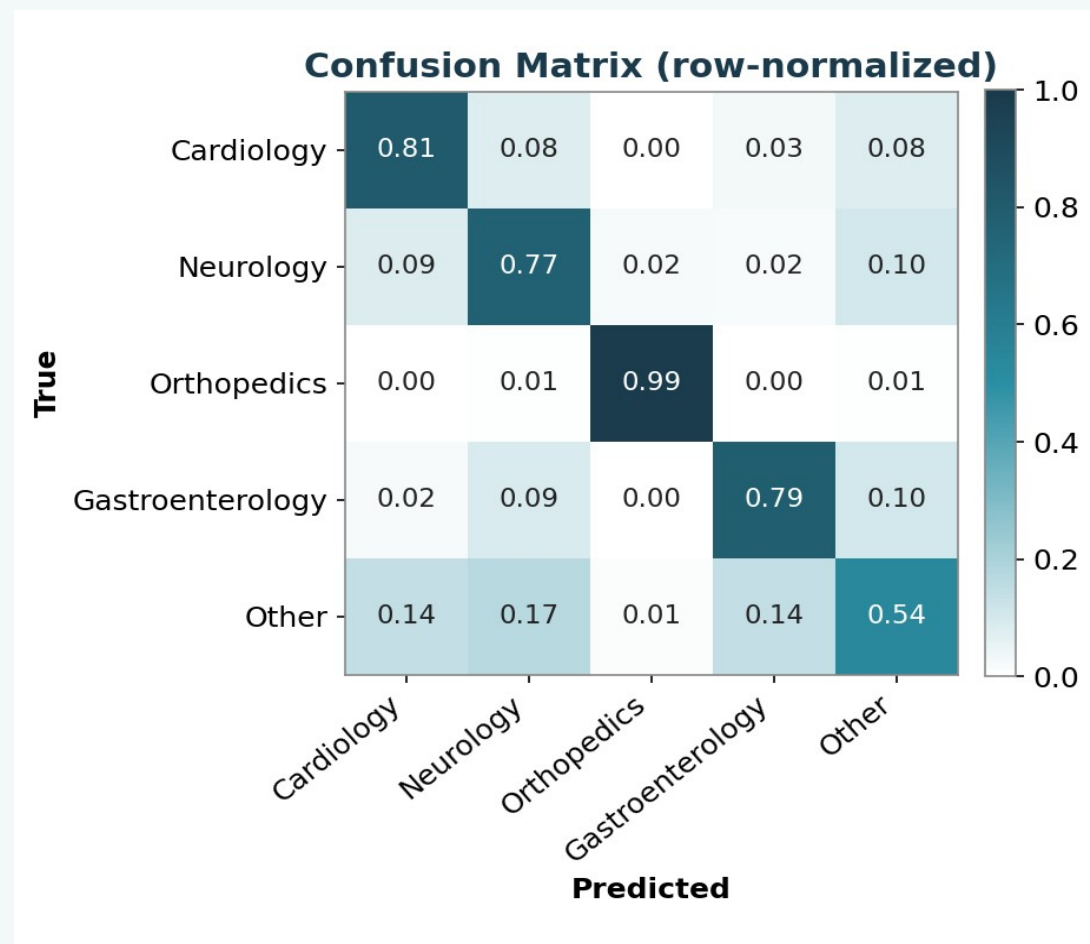
Macro-F1

0.93

Top-2 accuracy - how often the correct category is among the model's two highest-scoring guesses

Validated by 5-fold CV and an 80/20 holdout - the two agree, so no overfitting.

The four specialties are caught well (F1 0.58-0.97); Other is the bottleneck.



A calibrated probability per specialty

0.92

Macro ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) - how well the model gives the correct category a higher score than wrong ones

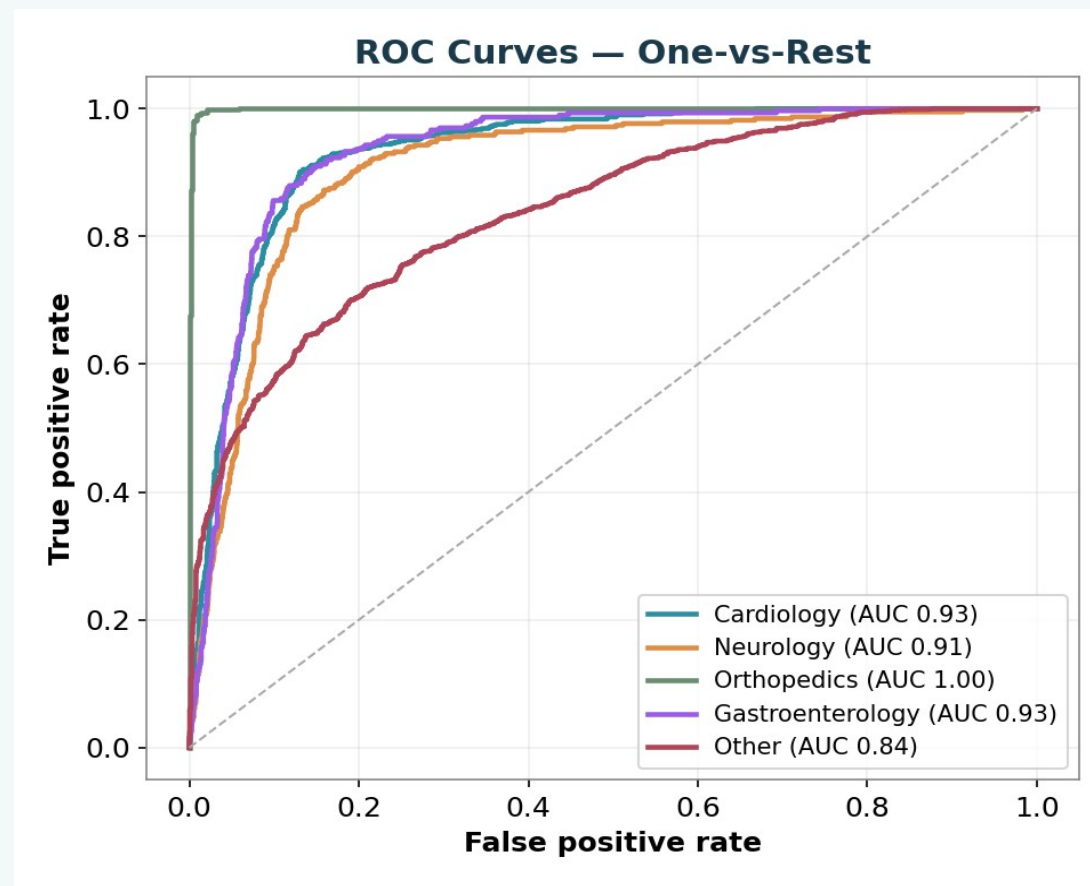
0.93

Top-2 accuracy - how often the correct category is in the model's top two guesses

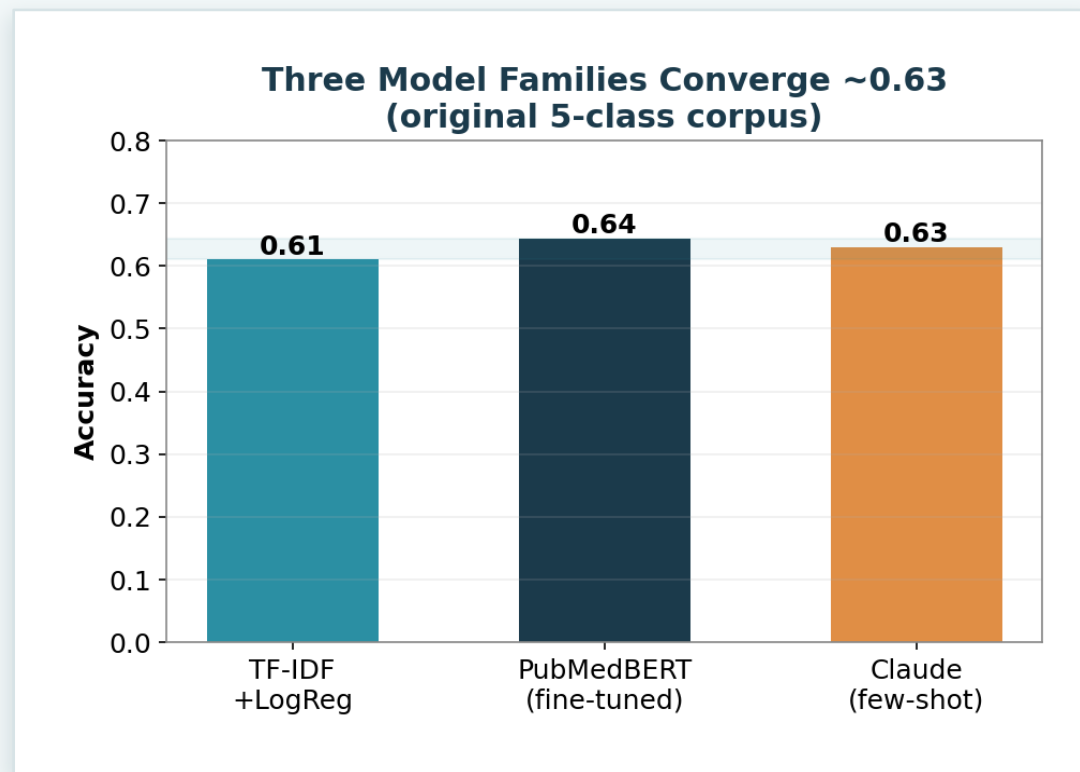
0.90

Log loss - how wrong the model's probabilities are; lower is better

Why probabilities? Two close top scores flag a note that spans departments (possible comorbidity); a flat distribution flags garbage to abstain on.



Three families converge - the ceiling is the data



Linear, transformer, and LLM all land ~0.6-0.7 and fail on the same catch-all class - strong evidence the limit is label overlap, not the model.

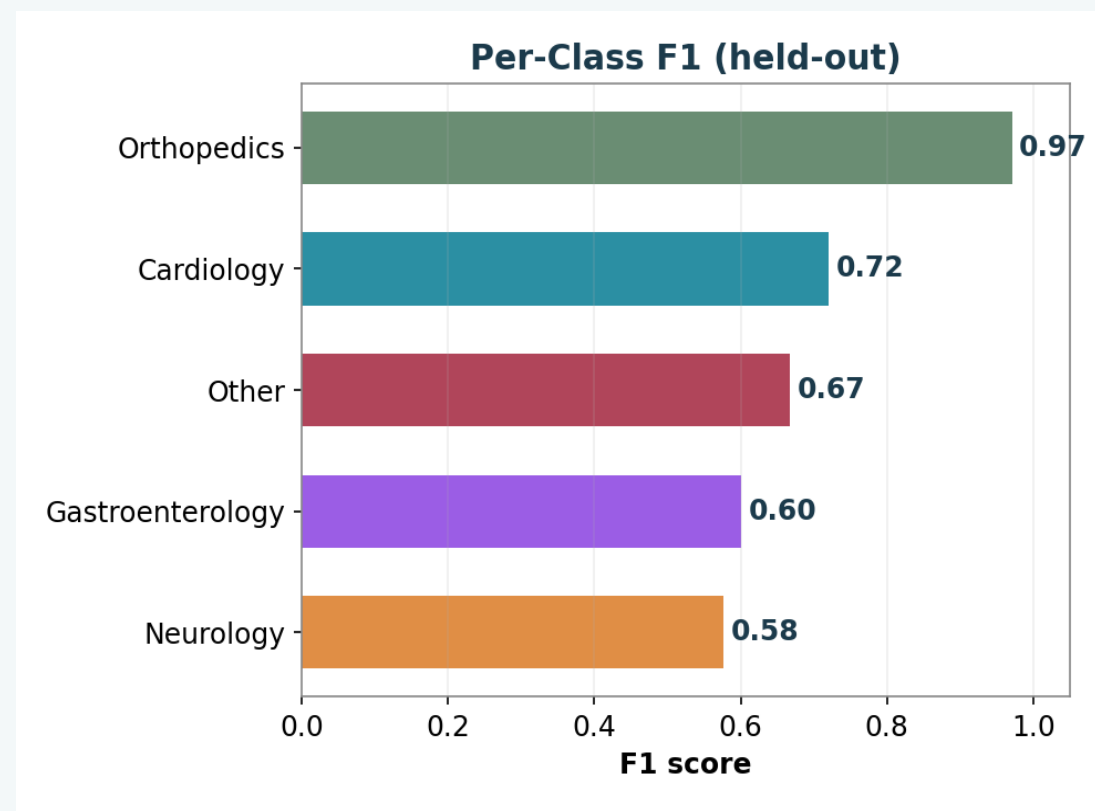
Why ~0.70, and why higher is hard

“Other” is a grab-bag

Oncology + general + everything else. It overlaps every class, so notes scatter into the four specialties (recall ~0.54).

Vocabulary vs meaning

Specialties separate by distinctive terms; the residual errors are genuine ambiguity no model resolves.



Tested on 1,000 unseen clinical vignettes

A real domain shift

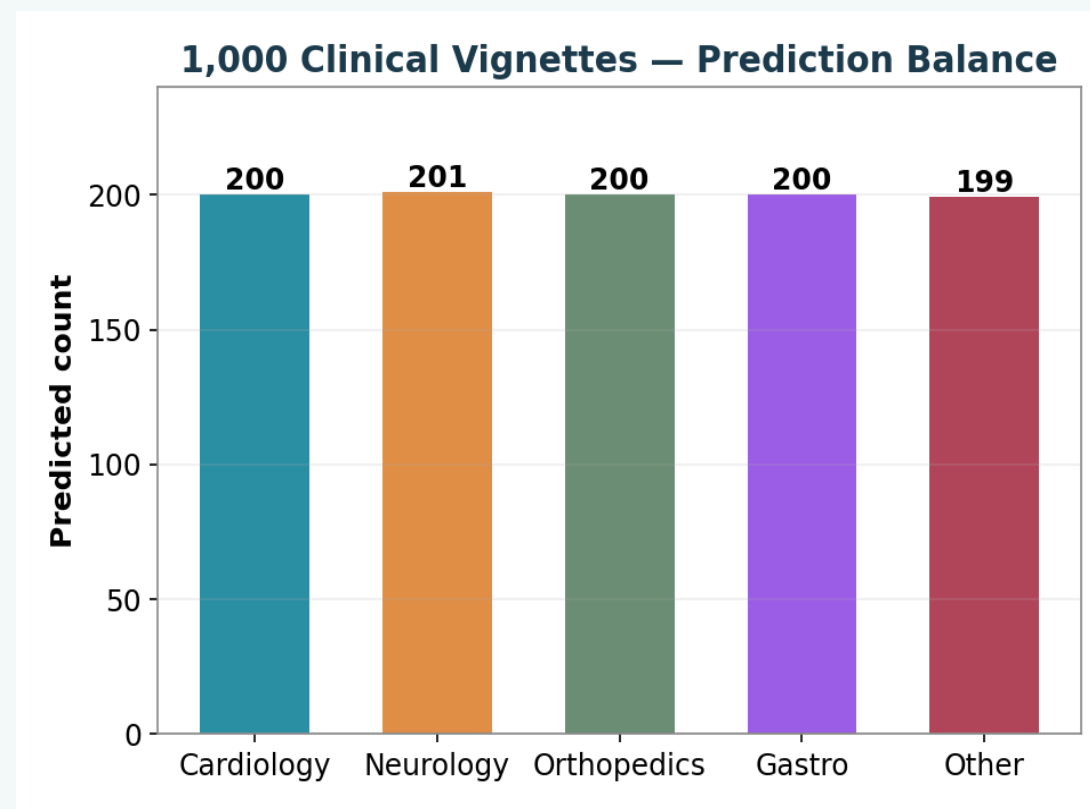
The test cases are short clinical transcripts - a different style from the training abstracts.

Semantic model transfers

The LLM classifies by meaning, not vocabulary, so it holds up where **bag-of-words** would degrade.

Balanced output

Predictions came out ~200 per class - a strong signal of accurate, unbiased transfer.



What we built

Models & Data

- Remapped the corpus to the 5 target categories
- Sourced an Orthopedics class from PubMed
- TF-IDF + LogReg submission (0.70 acc / 0.71 F1)
- Fine-tuned PubMedBERT transformer
- Claude LLM classifier (few-shot, semantic)

Evaluation & Insight

- Single-class + multi-class probability views
- Full healthcare metric suite + calibration
- Error analysis: the catch-all class is the ceiling
- Generalization test on 1,000 vignettes
- Classified the test set, balanced ~200/class

Code for Better

Honest models, measured limits, a working classifier for all five specialties.

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